NFCAMPUS

A DIGITAL STUDENT IDENTIFICATION & ACCESS SYSTEM USING NFC TECHNOLOGY FOR ANDROID

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BACKGROUND & PROBLEM STATEMENT

Our college uses RFID access cards for entry into campus buildings, libraries, and labs.

PROBLEM

While effective, this system has a major flaw: "It relies solely on a physical card".

- Students are frequently locked out due to forgotten, lost, or damaged cards
- Leading to missed classes, exams, and significant frustration.

VS

SOLUTION

The core issue is that the system identifies the card, not the person.

- Our solution leverages NFC technology on smartphones
- Creating a digital credential that integrates access control into a device students always carry.

OBJECTIVES

Develop a secure and reliable smartphone-based authentication system that supplements the existing college access card system.

SMART Criteria:

Specific	Develop a functional mobile application prototype that can emulate the college access card's credentials through NFC
Measurable	Prototype should successfully interact with test reader and trigger the "unlock" signal
Achievable	Use Android Studio and Host Card Emulation (HCE) for Android phones
Relevant	Provides a practical, modern supplement to the existing card system
Time-Bound	Project deadline: 28th March 2026

METHODOLOGY

Framework:
Agile-Scrum approach with iterative development

- **Sprints**: Two-week development cycles
- **Sprint Planning**: Persistent hemolysis markers
- **Daily Stand-ups**: Synchronize activities and identify blockers
- **Sprint Review**: Demonstrate completed functionality
- **Retrospective**: Process improvement and refinement

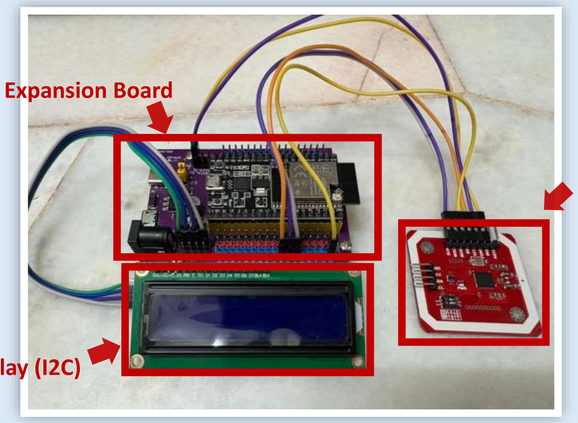
IMPLEMENTATION PLAN

Module 1: Basic Foundations
- Develop a backend server, database, and basic API endpoints
- Build an Android app with a login system, user interface and navigation

Module 2: Core Features
- OCR registration flow, NFC UID capture
- Password and email login, one-card-per-account system

Module 3: HCE Integration
- Implement HCE service to emulate card's UID
- Integrate "Simulate Access" feature

Module 4: Testing & Polishing
- End-to-end testing, security auditing
- User testing, documentation



TESTING & EVALUATION

Unit Testing
- Test each part of the system (e.g., Login, NFC Scan, Database)
- Make sure every module works properly before integration

Integration Testing
- Combine all modules and test how they work together
- Check communication between the app, server, and NFC reader

Security Testing
- Check how well the system protects user data and access
- Test for unauthorized access, NFC cloning, and encryption

System Testing
- Test the full system in real-world scenarios
- Try cases like valid/invalid users, lost devices, and no internet

Usability Testing
- Let students use the app and give feedback
- Measure ease of use, speed, and satisfaction

Performance Evaluation
- Evaluate the system's speed and reliability under heavy use
- Simulate multiple users and long usage periods

KEY FEATURES

- **User Authentication:** Email and password login system
- **OCR Integration:** Automatic card scanning
- **NFC UID Binding:** Secure card-to-account linking
- **HCE Technology:** Smartphone card emulation
- **Cloud Backend:** Firebase-powered data management
- **Secure Storage:** Encrypted credential protection

PROJECT TIMELINE

Phase 0: Initial Setup & Exploration
Timeline: Sept 2025
- Environment setup (Android Studio, GitHub)
- POC: Read NFC UID on test device
- Requirement gathering & stakeholder interviews

Phase 1: Research & Proposal
Timeline: Sept – Oct 2025
- Literature review & competitive analysis
- Define scope, SMART objectives & methodology
- Proposal writing, review & submission

Phase 2: System Design & Architecture
Timeline: Oct – Nov 2025
- System architecture & module planning
- UI/UX wireframes (Figma)
- Database schema & NFC/OCR flow design

Phase 3: Core Feature Development
Timeline: Nov – Dec 2025
- Login & registration module
- OCR for student ID
- NFC UID binding & login

Phase 4: HCE Integration & Prototype
Timeline: Jan – Feb 2026
- Host Card Emulation (HCE) implementation
- Simulated access / demo functionality
- Prototype validation

Phase 5: Testing & Improvements
Timeline: Feb – Mar 2026
- Unit, integration & security testing
- Usability testing & performance tuning
- Bug fixes & refinements

Phase 6: Documentation & Final Submission
Timeline: Mar 2026
- Update documentation
- Final report & documentation
- Demo, presentation & submission

EXPECTED DELIVERABLES

Category	Deliverable
Application	Fully functional Android app with OCR, NFC, and HCE
Backend	Firebase system with user accounts and UID validation
Documentation	System design, UML diagrams, API documentation
Testing	Test plans, reports, and security audits
Presentation	Final slides, demo video, project report

EXPECTED OUTCOME & SIGNIFICANCE

Anticipated Results:
- A fully functional Android application
- Seamless NFC-based access control
- Enhanced security model

Contributions:
Academic: Advancement in mobile computing and security
Industrial: Modernization of campus infrastructure
Societal: Enhanced user convenience and inclusivity
Beneficiaries: Students and the academic institution

Android Studio | Firebase | Cloud Firestore | Kotlin | Node.js | Javascript | GitHub | Figma

SCAN HERE FOR REFRENCES